PUNE INSTITUTE OF COMPUTER TECHNOLOGY

DHANKAWADI, PUNE –43

Class: B.E. **Semester:** I

Schedule of Assignments

Subject: Laboratory Practice IV (410247)

EXP. NO.	PROBLEM STATEMENT	Date of Completion
410244 (C): Cyber Security and Digital Forensics		
Group 1		
1.	Write a program for Tracking Emails and Investigating Email Crimes. i.e. Write a program to analyse e–mail header.	17 Jul 2026
2.	Implement a program to generate and verify CAPTCHA image.	24 Jul 2026
3.	A person on a nearby road is trying to enter a WiFi network by trying to crack the Password to use the IP Printer resource; write a program to detect such an attempt and prohibit the access. Develop the necessary scenario by Using an IEEE 802.11, configure a Wi-Fi adapter and Access Point.	31 July 2026
4.	Write a computer forensic application program for Recovering permanent Deleted Files and Deleted Partitions.	14 Aug 2026
5.	Write a program for Log Capturing and Event Correlation.	21 Aug 2026
Group 2 **(01 Mini- project)**		
1.	**Mini–project-** Design and develop a tool for digital forensic of images.	9 Oct 2026
2.	**Mini Project -** Design and develop a tool for digital forensic of audio.	
3.	**Mini Project -:** Design and develop a tool for digital forensic of video.	
4.	**Mini Project** - Design a system for the analysis of cyber-crime using various cyber forensic techniques and compare each technique with respect to integrity, confidentiality,	

	availability.	
410245(A): Information Retrieval		
Group 1		
1.	Write a program to Compute Similarity between two text documents.	28 Aug 2026
2.	Implement Page Rank Algorithm.	04 Sep 2026
3.	Write a map-reduce program to count the number of occurrences of each alphabetic character in the given dataset. The count for each letter should be case-insensitive (i.e., include both upper-case and lower-case versions of the letter; Ignore non-alphabetic characters).	11 Sep 2026
4.	Write a program to implement a simple web crawler.	18 Sep 2026
5.	Write a program to parse XML text, generate Web graph and compute topic specific page	25 Sep 2026
Group 2 **(01 Mini- project)**		
1.	Mini project: Develop Document summarization system	16 Oct 2026
2.	Mini Project: Develop Tweet sentiment analysis system	
3.	Mini Project: Develop Fake news detection system	
4.	Mini Project: Develop a Abstractive summarization system	

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